Hongbo Zhang

Madison, WI | h06234809@gmail.com

SUMMARY

GIS master's student with hands-on experience in nationwide geospatial analysis, wildfire-related spatial modeling, and Python-based GIS automation using ArcGIS Pro, QGIS, and geospatial Python libraries.

EDUCATION

University of Wisconsin–Madison, WI Expected May 2026

M.S. in Geography (Cartography and Geographic Information Systems)

Bellevue University, NE Jun 2024

B.S. in Management (Supply Chain Management)

TECHNICAL SKILLS

GIS & Visualization: ArcGIS Pro, QGIS, Mapbox Studio, Adobe Illustrator

Programming: Python (rasterio, geopandas, GDAL, numpy), JavaScript (Mapbox GL JS), HTML/CSS

Data & Analysis: Spatial SQL, GeoJSON, Shapefile, Raster Processing, Spatial Analysis, Geodatabase, QA/QC, Network Analyst

EXPERIENCE

GIS Research Assistant Jun 2025 – Present

University of Wisconsin–Madison, WI

- Conducted nationwide Wildland–Urban Interface analysis using ArcGIS Pro and Python to compare WUI-P, WUI-S, and WUI-Z patterns across the conterminous United States.
- Processed multi-source nationwide geospatial datasets for 48-state Wildland–Urban Interface comparison, supporting cross-region quality control and map production.
- Prepared and validated spatial outputs for cross-region comparison, quality control, and map production.
- Integrated CDC health indicators with geospatial outputs to support health–environment relationship analysis at multiple geographic scales.

SELECTED GIS PROJECTS

California Seasonal Wildfire Risk Analysis

- Analyzed seasonal wildfire risk patterns in California using ArcGIS Pro by integrating wildfire hazard surfaces, forest cover, human activity extent, and emergency facility locations.
- Synthesized multi-source spatial datasets to evaluate how vegetation conditions, ignition patterns, and regional weather drivers influenced wildfire risk across northern and southern California.
- Produced maps and written findings that highlighted distinct regional drivers, including dry summer fuels and lightning-related fire activity in northern California and Santa Ana wind-driven fire conditions in southern California.

Madison Transit Accessibility and Bus Stop Optimization

- Evaluated transit accessibility in Madison using ArcGIS Pro, bus routes, stop locations, and Dane County demographic data.
- Applied Network Analyst location-allocation and 10-minute walking service area analysis to identify underserved areas in the existing bus stop network.
- Ranked 200 candidate bus stop locations using population- and income-based demand indicators to support equitable transit planning and service expansion recommendations.